Supplement S7: Question selection and answer-budget sensitivity

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1 Scope, methods and provenance

These exploratory analyses reuse the same 600 previously examined E3 participants from IPIP and 8,320 fitting people. Each fitting person supplies all 60 target responses for supervised question selection. The answer budget counts observed history answers from an evaluation person, not total training supervision. Order construction and decoder tuning are repeated within training folds; E3 is excluded from these fits but has influenced the research direction. This is retrospective masking of complete questionnaires, not prospective administration or independent confirmation.

Scalar target selection centers history inputs and forms the 240 cumulative target indicators. With $G = X^T X + 100I$ and $C = X^T T$, it greedily maximizes $\|C_i\|^2/G_{ii}$ and updates residual quantities by a Schur complement. The selection penalty is distinct from the decoder penalty, retuned at each budget. Strong comparisons use the historical prior/ridge/kNN/ExtraTrees library; target-directed forms select ridge. Historical raw-answer ridge already uses one-hot inputs and isotonic CDF projection. The separate encoding ablation uses common sorted/clipped CDFs. Its gain must not be added to the strong comparison.

In the fixed-form contrast labels, "raw" denotes the scalar-encoding ablation and "category" denotes one-hot encoding; "strong" denotes the historical candidate library, whose raw-answer ridge already uses one-hot encoding. These labels must not be confused with the scalar representation used to learn the question order in the main strong comparison.

The 32 random-form tasks 28416_1–32 completed; they are forms on common people, not independent replications. Grouped-selector job 28444 completed after job28442 failed on a singular selected block; the correction excludes already selected blocks from the solve. The failed engineering receipt is retained. Curve job28457 completed. Exported grouped/curve aggregates were byte-matched to Euler results. This manuscript update performs no scientific rerun.

2 Fixed-form and mechanism contrasts

Differences are first method minus reference; negative favors the first method. Intervals below are pointwise, paired by participant and conditional on fixed fits and question sets. The original within-experiment Holm tests address zero differences, not a multiplicity-controlled practical-benefit claim. Full-precision estimates, p-values and original flags remain in the accompanying JSON. At five answers, target versus coverage has an interval whose upper bound is above -0.001; no useful-gain claim at that margin follows. Grouped and adaptive null/mixed outcomes are retained, without equivalence claims.

2.1 Fixed forms and encoding

Contrast	Δ RPS	CI low	CI high
m5/target_vs_random_raw	-0.00594	-0.00711	-0.00474
m5/target_vs_coverage_raw	-0.00167	-0.00268	-0.00074
m5/category_vs_raw_random	-0.00118	-0.00152	-0.00087
m5/category_vs_raw_target	-0.00157	-0.00205	-0.00109
m5/target_strong_vs_random_strong	-0.00632	-0.00766	-0.00498
m20/target_vs_random_raw	-0.00656	-0.00734	-0.00579
m20/target_vs_coverage_raw	-0.00212	-0.00269	-0.00156

Contrast	Δ RPS	CI low	CI high
m20/category_vs_raw_random	-0.00265	-0.00309	-0.00222
m20/category_vs_raw_target	-0.00357	-0.00410	-0.00305
m20/target_strong_vs_random_strong	-0.00750	-0.00839	-0.00665

2.2 Grouped versus scalar acquisition

Contrast	Δ RPS	CI low	CI high
m5/grouped_vs_scalar	0.00026	-0.00077	0.00131
m5/grouped_vs_marginal	-0.00227	-0.00314	-0.00138
m5/scalar_vs_marginal	-0.00253	-0.00349	-0.00158
m20/grouped_vs_scalar	0.00022	-0.00020	0.00065
m20/grouped_vs_marginal	-0.00537	-0.00623	-0.00451
m20/scalar_vs_marginal	-0.00559	-0.00648	-0.00471

2.3 Adaptive selection controls

Contrast	Δ RPS	CI low	CI high
m5/adaptive_vs_fixed	0.00055	-0.00021	0.00131
m5/adaptive_vs_random_same_decoder	-0.00480	-0.00579	-0.00381
m5/adaptive_vs_archived_strong	-0.00206	-0.00309	-0.00103
m5/adaptive_vs_fixed_target_strong	0.00426	0.00315	0.00537
m20/adaptive_vs_fixed	-0.00003	-0.00059	0.00055
m20/adaptive_vs_random_same_decoder	-0.00155	-0.00219	-0.00089
m20/adaptive_vs_archived_strong	0.00741	0.00650	0.00832
m20/adaptive_vs_fixed_target_strong	0.01492	0.01373	0.01612

3 Thirty-two additional random forms

Each random form retunes the same candidate library on FIT. No form has a smaller E3 point loss than the target-directed form at either budget. This rank statement is not a p-value or evidence from 32 independent samples. Target RPS is 0.13410 at five answers and 0.11699 at twenty. All per-form candidate selections are retained in random-forms.json.

Seed	RPS ($m = 5$)	RPS ($m = 20$)	Selected family (5/20)
1	0.13827	0.12174	ridge/ridge
2	0.14100	0.12302	ridge/ridge
3	0.14089	0.12393	ridge/ridge
4	0.13766	0.12325	ridge/ridge
5	0.14031	0.12195	ridge/ridge
6	0.13975	0.12543	ridge/ridge
7	0.13983	0.12350	ridge/ridge
8	0.14018	0.12258	ridge/ridge
9	0.13918	0.12299	ridge/ridge
10	0.13973	0.12505	ridge/ridge
11	0.13950	0.12239	ridge/ridge
12	0.14097	0.12404	ridge/ridge
13	0.14207	0.12516	ridge/ridge
14	0.14030	0.12298	ridge/ridge
15	0.14186	0.12577	ridge/ridge

Seed	RPS ($m = 5$)	RPS ($m = 20$)	Selected family (5/20)
16	0.13793	0.12257	ridge/ridge
17	0.13969	0.12509	ridge/ridge
18	0.13764	0.12299	ridge/ridge
19	0.13886	0.12158	ridge/ridge
20	0.14170	0.12362	ridge/ridge
21	0.14122	0.12428	ridge/ridge
22	0.13891	0.12159	ridge/ridge
23	0.13964	0.12166	ridge/ridge
24	0.14252	0.12516	ridge/ridge
25	0.13904	0.12439	ridge/ridge
26	0.14056	0.12527	ridge/ridge
27	0.14281	0.12556	ridge/ridge
28	0.13857	0.12624	ridge/ridge
29	0.13906	0.12323	ridge/ridge
30	0.14300	0.12498	ridge/ridge
31	0.13919	0.12378	ridge/ridge
32	0.14013	0.12290	ridge/ridge

4 Curve against the archived 60-answer Granite hybrid

All twenty examined budgets are reported. Positive differences favor the archived hybrid, whose mean RPS is 0.13153. The first negative mean difference is at seven answers, but its interval includes zero. The first interval entirely below zero is at eight; the first entirely below -0.001 is at nine. These are different criteria, not interchangeable crossing estimates. The historical random-order interpolation of 11.44 remains a separate result.

The stored curve uses 30,000 shared paired participant resamples with seed 20261115 and percentile endpoints 0.00125/0.99875 (Bonferroni across 20 cut points). These are approximate, conditional adjusted bootstrap intervals, not a guarantee of full-procedure coverage. The seed is an identifier, not an execution date. Fits, learned question order and historical hybrid recipes remain fixed.

Answers	Statistical RPS	Δ vs hybrid60	CI low	CI high
1	0.14668	0.01514	0.01348	0.01694
2	0.14232	0.01078	0.00905	0.01253
3	0.13920	0.00766	0.00595	0.00942
4	0.13682	0.00529	0.00359	0.00702
5	0.13410	0.00257	0.00079	0.00437
6	0.13254	0.00100	-0.00080	0.00283
7	0.13089	-0.00064	-0.00251	0.00126
8	0.12918	-0.00235	-0.00419	-0.00043
9	0.12741	-0.00413	-0.00595	-0.00229
10	0.12613	-0.00540	-0.00725	-0.00360
11	0.12496	-0.00657	-0.00841	-0.00476
12	0.12400	-0.00753	-0.00945	-0.00567
13	0.12299	-0.00854	-0.01045	-0.00667
14	0.12153	-0.01000	-0.01191	-0.00812
15	0.12058	-0.01095	-0.01286	-0.00913
16	0.11963	-0.01190	-0.01376	-0.01013
17	0.11906	-0.01247	-0.01433	-0.01068
18	0.11821	-0.01332	-0.01515	-0.01152
19	0.11751	-0.01402	-0.01586	-0.01220
20	0.11699	-0.01454	-0.01635	-0.01273

5 Boundaries of the comparison

The target-selected statistical predictor at five answers is worse than the 60-answer archived hybrid. At twenty it is better. Response budgets and total historical supervision differ: the statistical selector uses all target labels of 8,320 fitting people, while the hybrid has its own search, prior, selection and calibration history. These results do not measure LLM value at matched five or twenty answers or establish saved survey administrations. There are no new small-budget LLM forecasts. The separate equal60 audit remains unchanged.

The fitting-budget and facet-split diagnostics are retained as aggregates (fit-result.json and splits-result.json); their common participant cohort must not be counted as independent replications. No result here changes historical R2 Table1, its14 primary contrasts/686 endpoint rows, original Holm decisions, instruction comparisons or CAL-selection sensitivity. No new external panel is admitted and the A18 search stop is unchanged.